KINETIC THEORY OF GASES

Gas Processes & Their Graphs

Hold one of P, V, T fixed and the other two get locked together. Learn the three shapes once, and any cycle can be redrawn on any pair of axes.

◆ One equation behind every graph

$$PV = nRT$$

For a fixed amount of gas, n and R never change.

- **Isothermal** — T constant → $PV = \text{const}$
- **Isobaric** — P constant → $V \propto T$
- **Isochoric** — V constant → $P \propto T$

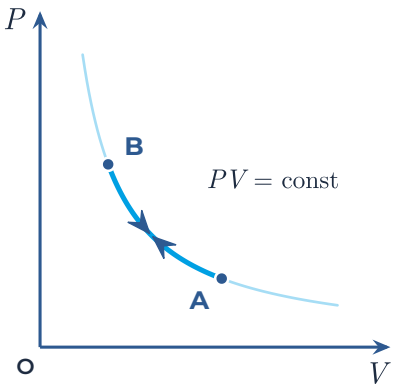
Each process keeps the same colour in every graph of these notes.

1 Isothermal Process

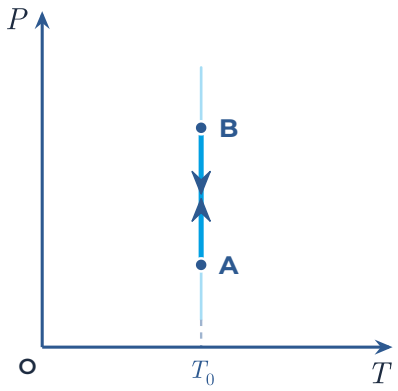
T = constant

$PV = nRT$ — n, R, T all constant $\Longrightarrow$ $PV = \text{constant}$ — *Boyle's law*

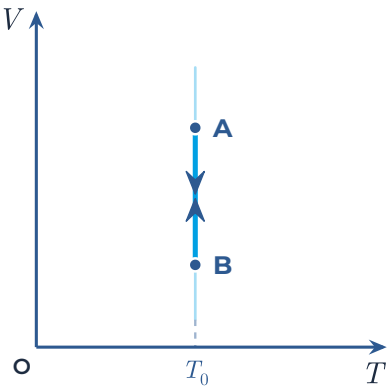
EXAMPLE Push a piston in **slowly** while the cylinder sits in a large water bath. Heat flows out into the bath, so T never changes.



P–V · rectangular hyperbola



P–T · vertical line at T_0



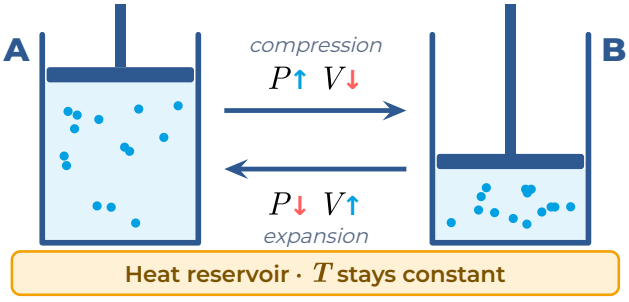
V–T · vertical line at T_0

A → B $P\uparrow$ $V\downarrow$ *compression*

B → A $P\downarrow$ $V\uparrow$ *expansion*

Read the arrows: the pair →← on a line means the gas can go either way, A → B or B → A.

Two isotherms? The hotter one ($PV = nRT$ larger) lies farther from the origin.

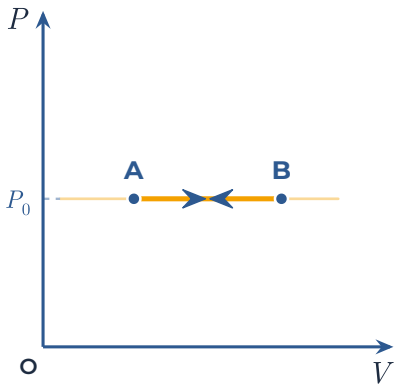


2 Isobaric Process

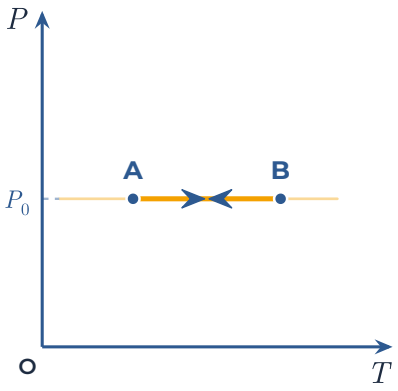
P = constant

$PV = nRT$ $\Longrightarrow V = \frac{nR}{P}\,T$ $\frac{nR}{P}$ *is constant* $\Longrightarrow V \propto T,\ V = kT$ *Charles' law*

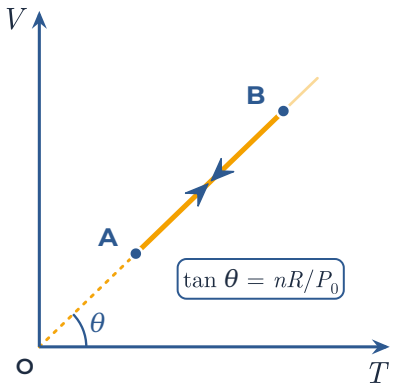
EXAMPLE Heat a gas under a **freely moving** piston. The piston rises and the gas expands, but the pressure on it stays the same.



P–V · horizontal line at P_0



P–T · horizontal line at P_0



V–T · straight line through O

A → B $V\uparrow\ T\uparrow$ *heated, gas expands* **B → A** $V\downarrow\ T\downarrow$ *cooled, gas contracts*

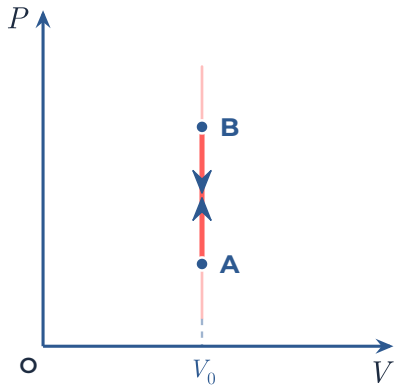
Slope: $\tan\theta = \frac{nR}{P_0}$. Lower pressure → steeper V–T line.

3 Isochoric Process

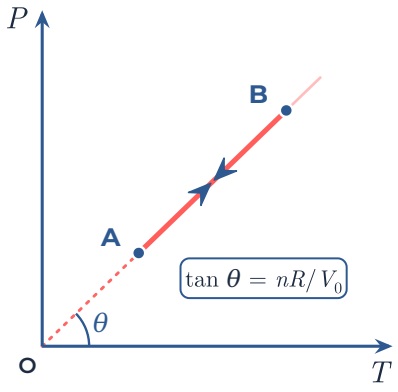
V = constant

$PV = nRT$ $\Longrightarrow P = \frac{nR}{V}\,T$ $\frac{nR}{V}$ *is constant* $\Longrightarrow P \propto T,\ P = kT$ *Gay-Lussac's law*

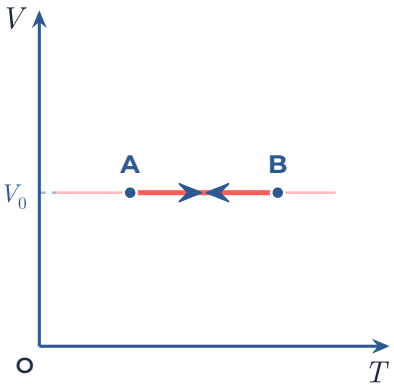
EXAMPLE Heat a gas in a **sealed rigid** container, like a closed pressure cooker. The volume cannot change, so the pressure climbs.



P–V · vertical line at V_0



P–T · straight line through O



V–T · horizontal line at V_0

A → B $P\uparrow\ T\uparrow$ *heated in a rigid vessel* **B → A** $P\downarrow\ T\downarrow$ *cooled in a rigid vessel*

Slope: $\tan\theta = \frac{nR}{V_0}$. Smaller volume → steeper P–T line.

At a Glance

Process	Constant	Relation	P–V graph	P–T graph	V–T graph
Isothermal	T	$PV = \text{const}$	Rectangular hyperbola	Vertical line	Vertical line
Isobaric	P	$V \propto T$	Horizontal line	Horizontal line	Line through O
Isochoric	V	$P \propto T$	Vertical line	Line through O	Horizontal line

RULE 1
The **constant** quantity on an axis gives a line **parallel to the other axis**.

RULE 2
Two quantities in **direct proportion** give a **straight line through O**.

RULE 3
$PV = \text{const}$ gives a **rectangular hyperbola** on the P–V graph.

Redrawing a cycle in 4 steps

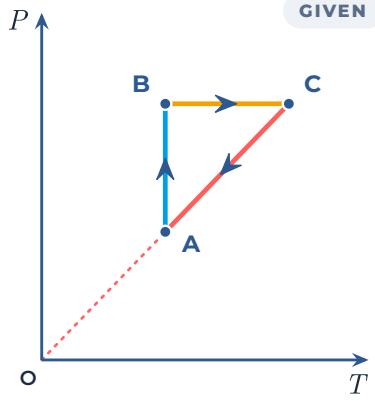
1. **Name each step.** Vertical or horizontal line → that quantity is constant. Line through O on P–T → V constant; on V–T → P constant.
2. **Write what rises and falls** in each step (P↑ V↓ ...).
3. **Place the points** on the new axes in the right order (e.g. $V_B < V_A$).
4. **Join them** with the shape from the table and keep the arrows in the same order A → B → C ...

Common slips to avoid

- An isothermal on the P–V graph is a **curve** (hyperbola), never a straight line.
- A straight line on P–T or V–T means isochoric / isobaric **only if it passes through O**. Extend it dotted to check.
- Keep the **same order of arrows** (A → B → C ...) on every graph you redraw.

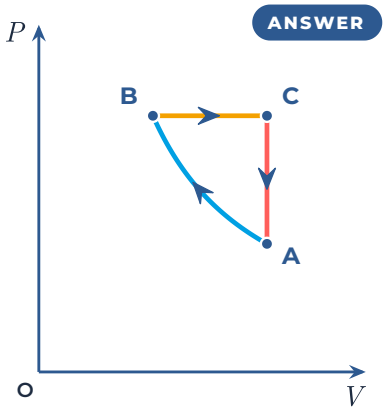
Q1 For the cycle **A → B → C → A** shown on the P–T graph, draw the P–V and V–T graphs.

GIVEN



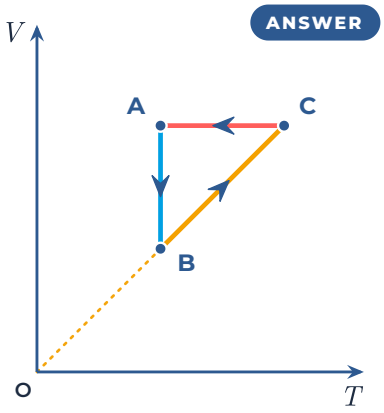
Given · P–T

ANSWER



P–V

ANSWER



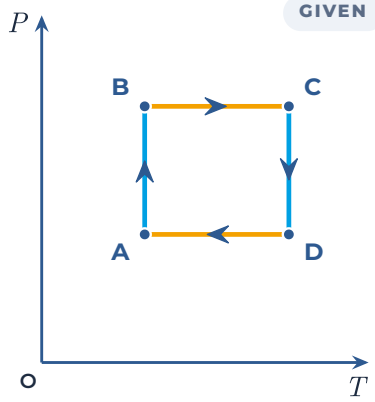
V–T

A → B	Isothermal	T const · P↑ V↓
B → C	Isobaric	P const · T↑ V↑
C → A	Isochoric	V const · P↓ T↓

KEY
C → A lies on a line through O on the P–T graph, so P/T is constant → V is constant. Hence $V_C = V_A$: a vertical drop on P–V, a horizontal line on V–T.

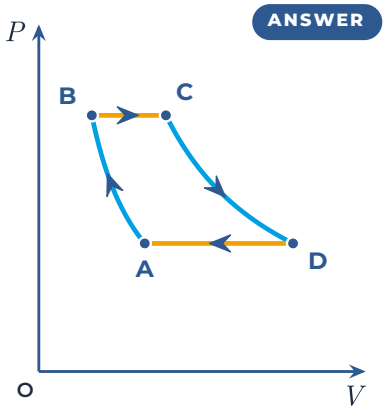
Q2 For the cycle **A → B → C → D → A** shown on the P–T graph, draw the P–V and V–T graphs.

GIVEN



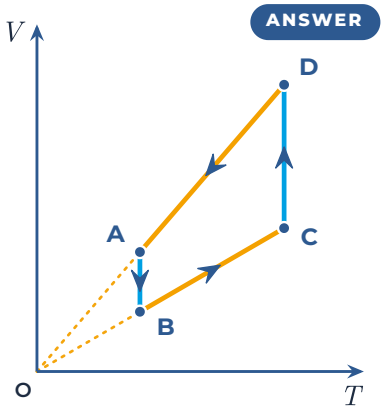
Given · P–T

ANSWER



P–V

ANSWER



V–T

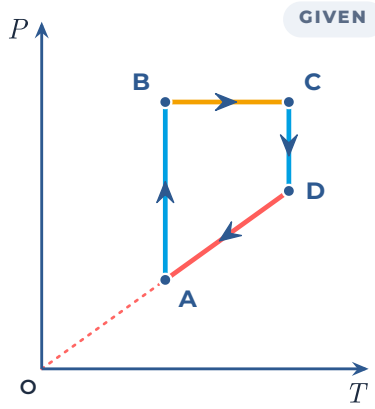
A → B	Isothermal	T const · P↑ V↓
B → C	Isobaric	P const · T↑ V↑
C → D	Isothermal	T const · P↓ V↑
D → A	Isobaric	P const · T↓ V↓

KEY

C → D is the hotter isotherm, so its hyperbola sits farther from O. On V–T, D → A is at the lower pressure, so its line through O is steeper than B → C.

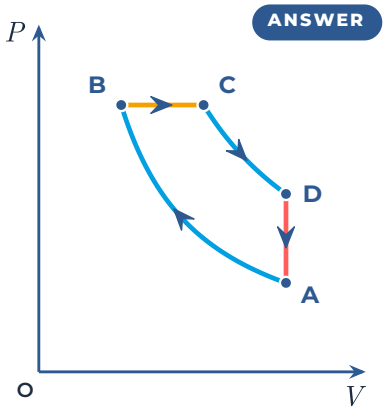
Q3 For the cycle **A → B → C → D → A** shown on the P–T graph, draw the P–V and V–T graphs.

GIVEN



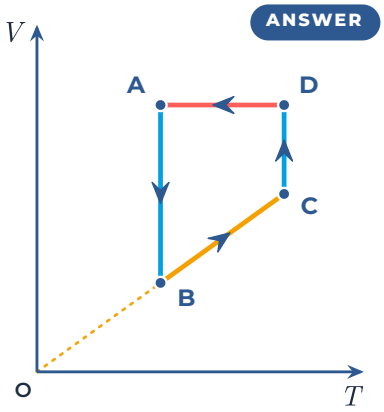
Given · P–T

ANSWER



P–V

ANSWER



V–T

A → B	Isothermal	T const · P↑ V↓
B → C	Isobaric	P const · T↑ V↑
C → D	Isothermal	T const · P↓ V↑
D → A	Isochoric	V const · P↓ T↓

KEY

D → A lies on a line through O on the P–T graph, so $V_D = V_A$. On P–V it is a vertical drop from D to A; on V–T it is a horizontal line.